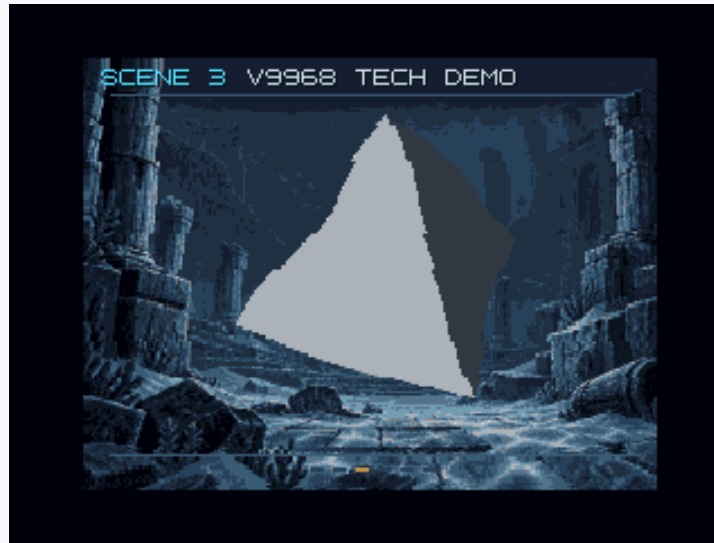


V9968 Scene 3 Optimization

About 1.5× faster in normal playback

Z80: 6.43 → 10.07 fps / R800: 8.24 → 12.31 fps



Measured in openMSX, FULL mode. Rendering optimized with pixel output preserved.

Normal playback: about 1.5× faster

FULL / Animated	Before	After	Speedup
C-BIOS / Z80	6.43 fps	10.07 fps	1.57×
FS-A1GT / R800	8.24 fps	12.31 fps	1.49×

Fixed poses benefit further from mesh caching and are evaluated separately.

Measurement conditions

All stages use the same Windows host, openMSX d884c4b, and z88dk/sdcc -SO3.

C-BIOS uses Z80; FS-A1GT uses R800 ROM mode. Display: 256×192, 16 colors.

Three consecutive 15-second intervals. $\text{FPS} = \text{completed frames} \times 60 / \text{elapsed VBlank ticks}$.

Includes HUD, PSG music and VSync waits. This measures emulated performance, not host speed.

P fixes mesh and water phase at zero. results-optimized.json records samples and ROM hashes.

FPS by stage: animated

Stage	C-BIOS / Z80	FS-A1GT / R800
Baseline	6.433	8.244
A: Remove full clear	6.548	8.539
AB: Work page + restore	6.716	8.867
ABC: Merge water runs	7.957	10.504
ABCD: Merge mesh rectangles	8.305	10.798
ABCDE: Assembly streaming	10.070	12.312

FULL mode. Frames and ticks pooled across three intervals. Raw samples and ROM hashes are in JSON.

FPS by stage: fixed pose

Stage	C-BIOS / Z80	FS-A1GT / R800
Baseline	6.000	7.500
A: Remove full clear	6.000	7.500
AB: Work page + restore	9.627	12.000
ABC: Merge water runs	12.000	15.000
ABCD: Merge mesh rectangles	12.000	15.000
ABCDE: Assembly streaming	13.005	20.000

FULL mode. Frames and ticks pooled across three intervals. Raw samples and ROM hashes are in JSON.

Why fixed poses improve more

Z80: 6.00 → 13.00 fps. R800: 7.50 → 20.00 fps.

Before optimization, every frame restored the background and redrew the mesh.

An unchanged mesh now skips restoration and drawing. Water, HUD and display updates continue.

Animated shapes and water phases advance with time, so cache reuse varies during playback.

VBlank waits quantize frame rates. Variation across consecutive intervals is not a confidence interval.

RGB5 / RGB3 and rendering speed

Animated	Z80 before → after	R800 before → after
FULL	6.43 → 10.07	8.24 → 12.31
FAST	6.44 → 10.07	8.32 → 12.32
COMPAT	1.95 → 3.87	2.09 → 4.09

FULL / RGB5



FAST / RGB3



RGB5: 32 levels per channel / RGB3: 8. Both display 16 colors at once.
Actual FULL and FAST captures at the same fixed pose; quantization changes dark and gray tones.

Scene 3: the four VRAM pages

Background 3 → clean composite 2 → water distortion + HUD → back page 1

Page 0: front



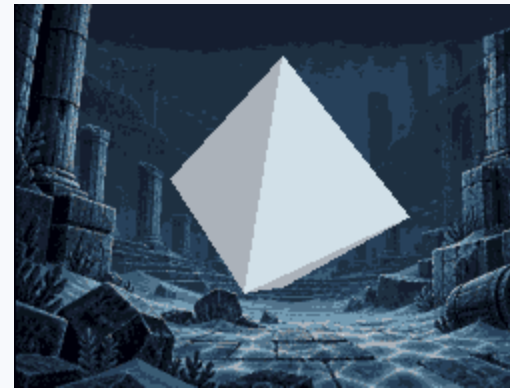
Currently displayed frame

Page 1: back



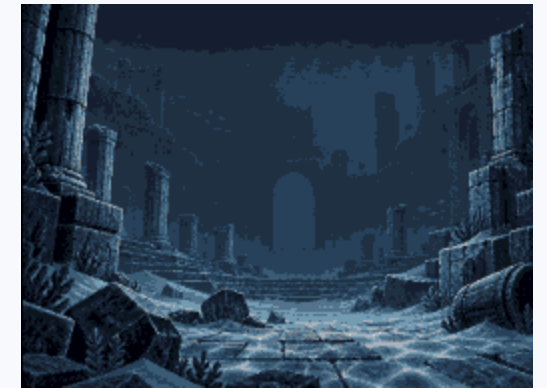
Completed next frame

Page 2: work



Background + mesh; no water/HUD

Page 3: source



Restoration source; no mesh

Swap 0/1 at VSync. On a new shape, restore its old bounds from page 3; otherwise reuse page 2. Page 3 also holds headers in rows 192-245, outside these 192-row captures.

Reduced transfer and drawing work

Work / average per frame	Before	After
Background restore pixels	49,152	15,298
Full clear + capture copy	98,304 pixels	0 pixels
Water commands	224.000	123.273
Mesh LMMV commands	257.141	223.836

When the shape changes. The 49,152-pixel water transfer and HUD remain. Same poses also skip mesh work.

Merge water bands without changing pixels

Merge only when horizontal mappings match and source/destination rows are contiguous.

Main transfers fall from 96 to 53.102 on average. Edge fills use the merged band height.

Repeat the outermost pixel twice; do not approximate it by copying a two-pixel pair.

Keep 512 bytes per phase: a count followed by five-byte sx, dx, width, sy, height records.

All 256 phases and 24,576 original bands checked against independent equations. Identity uses one copy.

Mesh rectangles and command streaming

Merge adjacent raster spans only when X, width and color match. All 128 images reconstruct exactly.

Keep 4,096 bytes per mesh frame. Store a four-byte bounding box at the end; no runtime scan.

Assembly accelerates CE polling and packet output. Protect R17 selection from interrupts.

Retain the 65,535-poll watchdog and fault stop. The ISR updates music/time without bank switching.

-CStream selects C mesh, glow and water paths. Compare rendered output; execution speed may differ.

Correctness and regression checks

Check	Result
Reconstruct all 128 original meshes	All pixels match
Water equations, 256 phases	PASS
Identity + edge pixels on both CPUs	Zero mismatched bytes
130 work-page captures + Scene 6 re-entry	All images match
Six scenes, headers, W, F/P/Esc, mapper	PASS

Standard V9958 COMPAT also passes on both CPUs. Injected timeout confirms the fault marker and display stop.

Before and after: image comparison

Before optimization



After optimization



Fixed pose, FULL, C-BIOS/Z80. The 256×160 image region excluding HUD matches exactly.

Implementation notes and validation scope

An in-place argument mask fails at -SO3; assign the masked result to a separate variable.

Tests include inputs 128–255 to verify correct selection of all 128 mesh frames.

FPS is measured in emulation. FPGA command timing, port spacing and audio remain unverified.

Rendered poses depend on FPS; pixel comparisons use matching poses and water phases.

The C path validates algorithms. Interrupt entry and atomic clock reads retain dedicated assembly.

Rebuild and repeat the measurements

Run from the repository root. Replace <z88dk> with your installation path.

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -File  
demos/scene3-benchmark/build.ps1 -Z88dk "<z88dk>"
```

```
powershell.exe -NoProfile -ExecutionPolicy Bypass -File  
demos/scene3-benchmark/measure-optimization.ps1 -Runtime runtime/cbios
```

For R800 use runtime/fsa1gt. Launch: launch.ps1 -Mode cbios -Optimized.

Add -CStream to test the C implementation. The standard distributed ROM uses assembly.

Measured ROMs and references

Baseline benchmark SHA-256

b96e3b0f3999e97d8c2b5c2a724f77921e5bf419b8b0febbba852be73e99421f8

Baseline rebuilt from commit be3bfde; not included in the distribution.

Optimized benchmark SHA-256

ce1504f2798d97ff56638db7f90ca72c79cdd8d14353d31d1ed97b61c39cc427

Optimized demo SHA-256

5f9577cf30c8dedbcf2ceffc70077beeabe002771726a7d944b6655d5bca8021

Demo hash is the 0.7.1 ROM; 0.7.2 changed the mesh source.

Details: OPTIMIZATION.md / OPTIMIZATION.ja.md, results-optimized.json, generators and tests.

Pinned VDP implementation: [buppu3/openMSX](#) d884c4b

[src/video/VDPCmdEngine.cc](#) / [VDP.hh](#) / [VDP.cc](#)